



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to CHENDUR FUEL SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

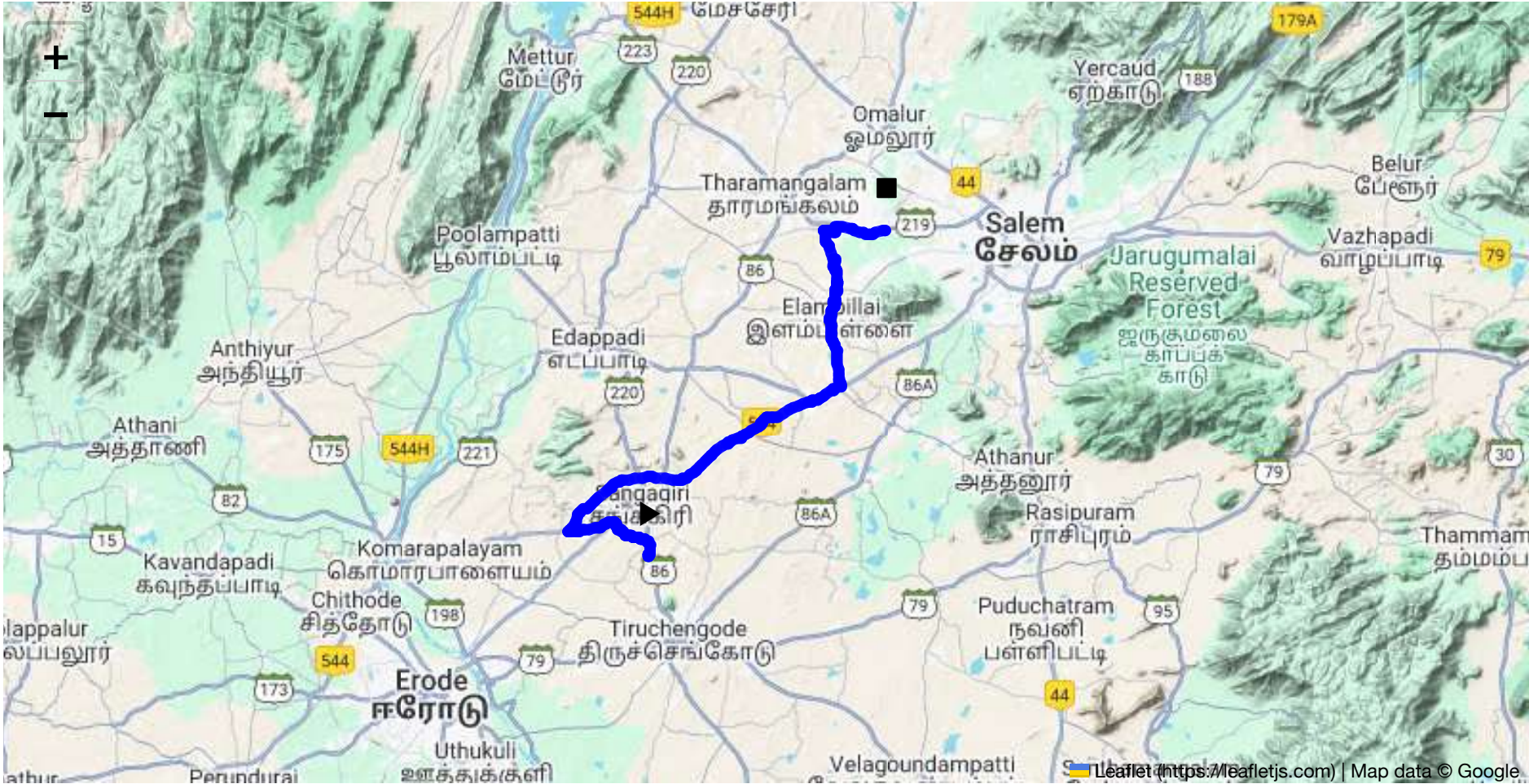
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 52.25 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours

Start: (11.4381, 77.8734)
End: (11.668705, 78.044135)



Welcome to the Journey Risk Management Study

Route Map Overview

The route between Sangagiri, Tamil Nadu, and Maramangalathupatti, Tamil Nadu via Bhavani Main Road in Padaiveedu and Omalur spans approximately 52.25 kilometers. Starting from CVQF+23W, you head towards M29V+JH3. The route takes you through rural and semi-urban areas, with a mix of state highways and local roads. Expect a drive time of approximately 1.21 hours under normal conditions for heavy vehicles carrying hazardous materials.

Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate with hot and dry weather for most parts of the year. The region can experience intense rainfall during the monsoon months (October to December). Driving in these conditions may lead to reduced visibility and slippery roads. Flash floods are a potential risk during severe storms.

Traffic Patterns

This route experiences moderate traffic, primarily because it passes through rural and connecting areas rather than major urban centers. However, traffic can get congested around market areas and town centers, particularly during morning and evening rush hours (8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM).

Road Quality and Infrastructure

The roads vary from well-maintained highways to narrower local roads. While state highways like Bhavani Main Road are generally in good condition, some local roads may have potholes and limited signage. Watch for cattle or pedestrians, especially in village areas.

Alternative Routes for Emergencies

In emergencies, you might consider the following alternative routes:

1. Instead of passing through Padaiveedu, continue on the State Highway directly if signage and infrastructure allow, or
2. Use nearby state or district roads as marked detours connecting to other highways to bypass congestion.

Local Regulations for Hazardous Material Transport

Local regulations typically require that hazardous materials be transported in compliance with safety protocols, including appropriate labeling and reporting to local authorities, especially when passing through populated areas. Check with local transportation offices for specific regulations.

Historical Incidents

There is limited public data on historical incidents involving heavy vehicles or hazardous materials on this exact route. However, general caution is advised due to scattered reports of accidents involving heavy trucks on rural roads due to overloading and driver fatigue.

Environmental Considerations

The route passes through agricultural areas sensitive to spills or contamination from hazardous materials. Vigilant adherence to safety measures is necessary to prevent environmental damage.

Communication Coverage

The communication network along the route is generally reliable, though some rural areas between larger towns might have weaker signals or intermittent coverage. It’s advisable to have an alternative communication method (e.g., offline maps, radios) in these zones.

Estimated Emergency Response Times

- In urban sections near Omalur and Maramangalathupatti: 15 to 30 minutes.
- In rural or less accessible sections: 30 to 60 minutes.

Overall Summary of Risk Assessment

The route presents a moderate-risk level for transporting hazardous materials. Key risks include weather-related disruptions, rural road conditions, and potential communication issues. Timely planning, especially during monsoon season, can mitigate risks. Continuous monitoring of traffic and weather conditions is recommended, along with compliance with local regulations to ensure safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45839, 77.85143	30 KM/Hr
6	Turn	Medium	11.46307, 77.84941	30 KM/Hr
7	Turn	Medium	11.46316, 77.84921	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
10	Turn	High	11.55839, 78.01288	15 KM/Hr
11	Turn	Medium	11.60526, 78.00451	30 KM/Hr
12	Turn	Medium	11.61685, 78.00830	30 KM/Hr
13	Turn	High	11.62660, 78.00986	15 KM/Hr
14	Turn	High	11.62692, 78.00883	15 KM/Hr
15	Turn	Medium	11.64031, 78.00923	30 KM/Hr
16	Turn	Medium	11.64874, 78.00717	30 KM/Hr
17	Turn	Medium	11.66065, 78.00106	30 KM/Hr
18	Blind Spot	Blind Spot	11.66821, 78.00077	10 KM/Hr
19	Turn	High	11.66762, 78.00863	15 KM/Hr
20	Blind Spot	Blind Spot	11.67168, 78.00922	10 KM/Hr
21	Blind Spot	Blind Spot	11.66830, 78.04357	10 KM/Hr
22	Blind Spot	Blind Spot	11.66847, 78.04352	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vembadithalam Government General Hospital	11.5683, 78.01211	30 km/h	Medium
4	hospital	Subam Hospital	11.6069532, 78.0042593	30 km/h	Medium
5	hospital	SR Hospital	11.6075985, 78.0070051	30 km/h	Medium
6	hospital	Nesam Medical Centre and Hospital	11.6078647, 78.0073456	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



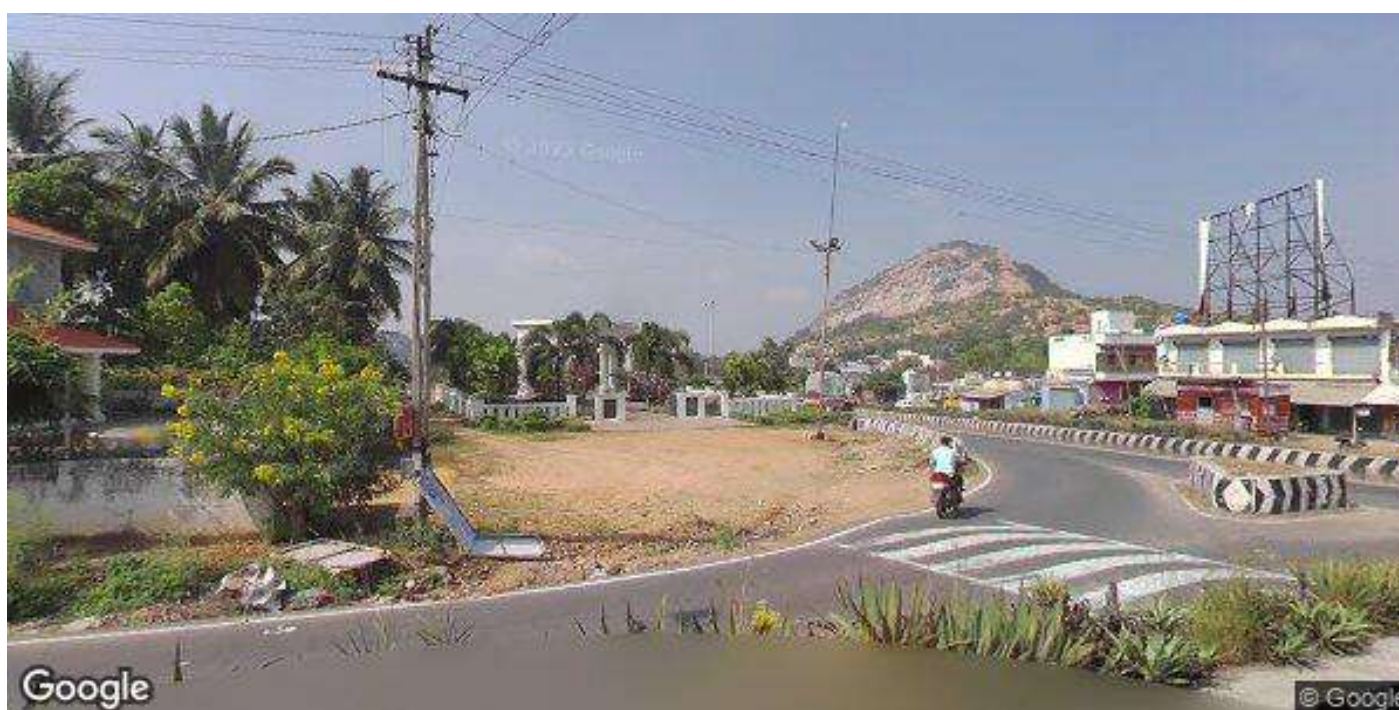
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60526, 78.00451



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62660, 78.00986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62692, 78.00883



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64031, 78.00923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64874, 78.00717



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.66821, 78.00077



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66762, 78.00863



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.67168, 78.00922



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.66830, 78.04357



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.66847, 78.04352

